Exercise 4

When is a scheduling method considered fair?: A scheduling algorithm is a one that gives each process CPU time, it prevents starvation and makes sure that each process gets some time. The CPU allocation does not have to be equal the waiting time does not have to be equal as so for the running time. It also define bound waiting time, for how long a process shall wait as max before getting some CPU time.

Some Examples of Fair Scheduling Algorithms: Fair share, Round Robin

What does preemptive mean in scheduling methods?: A preemptive scheduling method is one that can interrupt a process to give another process some CPU assets to run, it keeps pausing and running processes. It eventually simulates as the processes are running at the same time.

Which scheduling methods are preemptive?: Fair share, round robin, earliest deadline first.

Unix systems use priority-based scheduling in part, where the priorities of user processes can also be changed at runtime. Research which priority ranges exist under Linux and which of these are available to user processes.: Linux supports real-time priorities ranging from 0 to 99, which are reserved for privileged processes. Normal user processes use time-sharing priorities derived from nice values ranging from −20 to +19, corresponding to internal priorities 100 to 139. Users can lower their own priority but require special privileges to increase it.

Also find out how the priority of a user process can be set by the user at program start as well as how the priority can also be changed dynamically: A user in linux can assign a priority value when first initiating the program by adding a nice value of -20 – 19. Dynamically this can be changed using the renice.

Normally round roben algorithm does not care about the priority, just the ready first.